CS3AM Project Plan: Bankruptcy Early Warning

1. Overview

Working title: **Can machine learning beat the Altman Z′ score? Predicting corporate bankruptcy one year ahead from financial ratios.**

Lenders, credit insurers and auditors must judge whether a firm will fail before extending credit or signing off accounts, and a missed bankruptcy costs far more than a false alarm. Practice still leans on linear scores such as Altman's Z′ (Altman, 1983). The project measures how much modern models add over that benchmark.

Research questions

1. Do XGBoost and a neural network rank firms by one year bankruptcy risk better than the original Altman Z′, a Z′ refitted on the Polish data, and a logistic regression on all 64 ratios?
2. Does the neural network add anything over gradient boosted trees on tabular ratio data?
3. How fast does predictive power decay as the horizon lengthens from 1 year to 5 years, measured on ROC AUC, with PR AUC divided by test prevalence as supporting evidence?

Hypotheses

- **H1a, H1b, H1c:** XGBoost and the MLP each beat (a) the original Z′, (b) the refitted Z′ and (c) logistic regression on 64 ratios, on test PR AUC. That is six comparisons; each passes if the paired bootstrap 95% interval of the PR AUC difference (model minus comparator) excludes zero on the positive side. All six intervals are reported, with no pooled verdict and no multiple comparison correction. A failed hypothesis is reported as a finding.
- **H2:** XGBoost matches or beats the MLP (Grinsztajn et al., 2022), with a margin of 0.05 PR AUC. With about 82 test positives, the paired bootstrap interval of a PR AUC difference is expected to span roughly ±0.05, so a tighter margin could not be met even by two equally good models. Verdicts, checked in this order: interval entirely above zero, XGBoost better; interval entirely below zero, MLP better; lower bound above −0.05, XGBoost matches; otherwise inconclusive. H2 passes on "XGBoost better" or "XGBoost matches".

- **H3:** both models score higher on the 1 year horizon file (`5year.arff`) than on the 5 year horizon file (`1year.arff`). Decision metric: ROC AUC; PR AUC divided by test prevalence is supporting. For each model, H3 passes if the 95% interval of (metric on `5year.arff` minus metric on `1year.arff`), from independent stratified bootstraps of the two test sets, lies above zero. The five file trend is described, not tested.

Fixed in advance. Hypothesis rules, margins, the nominated seed (42) and the Platt choice are committed to `docs/decision_log.md` by Gate 1 and not altered after the test run.

In scope: the five UCI files, 64 accounting ratios, XGBoost, a PyTorch MLP, the original Altman Z′, a refitted Z′ and logistic regression as reference points, SHAP explanations.

Out of scope: market prices, news text, macroeconomic data, causal claims, any deployed application.

2. Deliverables and rules

One zip on Blackboard by Fri 4 Dec 2026 at noon, submitted by Abdalla on Thu 3 Dec; the coursework is worth 50% of CS3AM.

- **Zip contents:** report (PDF or Word), the dataset files, Python scripts (PY or IPYNB) with comments.
- **Report length:** 5 to 7 pages, Times New Roman 12 pt, 1.15 spacing. Figures and tables count. Front page, references, the signed reflection table and appendices do not.
- **Front page fields:** module code, title, date, actual hours spent, other learning aids used.
- **Models:** two, one of them deep learning in PyTorch or TensorFlow.
- **GenAI:** allowed; listed under other learning aids. Borrowed code is referenced in comments.
- **Effort table:** each member signs an effort percentage for four categories (data, modelling, evaluation, report). The group mark is scaled down for anything below 100%. Everyone works in all four categories.

Where the marks are: about 45 of 100 marks reward justification, evaluation design, discussion and writing; 10 reward implementation. Every choice in the code needs a reason that can be written down.

3. Dataset

The main experiment uses `5year.arff` from the UCI Polish Companies Bankruptcy dataset: 5,910 firm statements, 410 bankrupt within one year (6.9%), published raw counts. Licence CC BY 4.0; cite Tomczak (2016), DOI 10.24432/C5F600. Download: zip, 8.4 MB.

File	Bankrupt within	Statements (raw)	Bankrupt (raw)	Bankrupt (%, raw)	Use
5year.arff	1 year	5,910	410	6.9	Main experiment (E1); its frozen split and E1 test results are reused in E2
1year.arff	5 years	7,027	271	3.9	Horizon experiment (H3)
2year, 3year, 4year	4, 3, 2 years	10,173 / 10,503 / 9,792	400 / 495 / 515	3.9 / 4.7 / 5.3	Horizon experiment (H3)

The table shows published raw counts. Counts after duplicate removal (applied to all five files) and the class weight (negatives divided by positives in the training split) are computed and logged by the code, never hard coded.

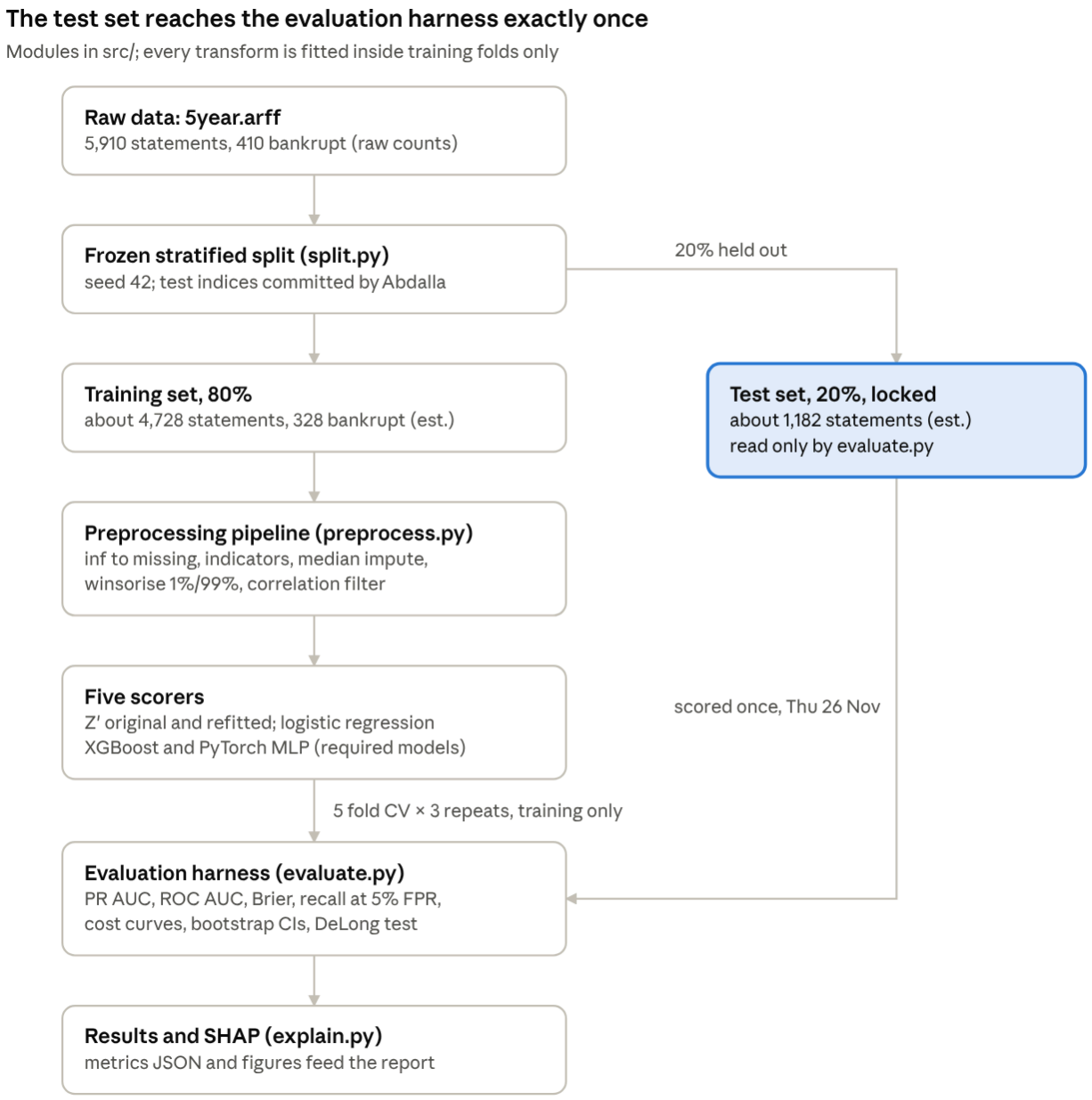
- File numbering runs opposite to the horizon: `1year` is the longest.
- Features: 64 ratios `Attr1` to `Attr64` (profitability, leverage, liquidity, activity, size) and `class` (1 = bankrupt).
- Altman Z′ inputs: Attr3 (working capital / total assets), Attr6 (retained earnings / total assets), Attr7 (EBIT / total assets), Attr8 (book equity / total liabilities), Attr9 (sales / total assets).

```
Z' = 0.717\,X_1 + 0.847\,X_2 + 3.107\,X_3 + 0.420\,X_4 + 0.998\,X_5
```

Known issues: missing values; infinite and extreme ratios from near zero denominators; 6.9% positives; bankrupt firms sampled 2000 to 2012 and survivors 2007 to 2013 (a limitation for the report); no firm identifiers, so files are never pooled; exact duplicate rows are removed by load.py before the split, with the count logged; the `ucimlrepo` package does not match the per file counts, so the zip is the only source.

4. Method

One frozen split, one preprocessing pipeline fitted inside training folds only, five scorers, and one evaluation harness that every model passes through.



Modelling pipeline · 8 stages, one locked test path

Three rules that are never broken

1. The test set is scored once, on Thu 26 Nov, by Abdalla (or Cameron as authorised deputy, handover logged), after the model freeze.
2. Every transform, calibrator and cost threshold is fitted on training data only (inside folds or on out of fold predictions), never on validation folds being scored or on test data.
3. The five horizon files are never pooled.

Split and validation. `load.py` removes exact duplicate rows from each of the five files and logs the count before any split. Stratified 80/20 split, seed 42; split sizes in this plan (about 4,728 training and 1,182 test statements, about 82 test bankruptcies) are estimates

before duplicate removal, and the actual counts are logged by the code. Tuning uses stratified 5 fold cross validation repeated 3 times on the training set. Inside CV, early stopping uses a stratified inner validation split (15%) of the training fold, never the fold being scored. For the final refit on the full training set, tree and epoch counts are fixed at the median best iteration from CV.

Bootstrap. 2,000 resamples, stratified by class, with the same resample indices across all scorers and the seed taken from `config.yaml`.

Preprocessing (sklearn `Pipeline`)

1. Infinite values to missing.
2. Missingness indicator for each ratio with more than 5% missing.
3. Median imputation.
4. Winsorise at the 1st and 99th percentiles of the training fold.
5. Drop one of each pair with absolute Spearman correlation above 0.95, keeping the less missing ratio. The five Altman inputs (Attr3, Attr6, Attr7, Attr8, Attr9) are never dropped; their correlated partner is dropped instead.
6. Both logistic regressions: standardise. MLP: standardise (or quantile transform if skew remains; decided on CV evidence).

Scorers

Scorer	Role	Key settings	Owner
Original Altman Z′	Finance benchmark, published coefficients	Computed on pipeline output (infinite to missing, median imputation, winsorising, all fitted on training data) so no row lacks a score; −Z′ as risk score	Abdalla
Refitted Z′	Benchmark re-estimated on the Polish data	Logistic regression on Attr3, Attr6, Attr7, Attr8, Attr9 only (`baselines.py`); standardised inputs; L2; class weight balanced; C tuned over 0.01, 0.1, 1, 10 with the same CV	Abdalla
Logistic regression	Statistical baseline on all 64 ratios	Standardised inputs; L2; class weight balanced; C tuned over 0.01, 0.1, 1, 10 with the same CV	Abdalla

Scorer	Role	Key settings	Owner
XGBoost	Model 1	Class weighting tuned: scale_pos_weight = negatives / positives in the training split (≈ 13.4, estimate before duplicate removal; computed and logged by the code) or 1; early stopping after 50 rounds on the inner split; final refit at median best iteration; 40 random configurations; final model trained on the 5 seeds in `config.yaml`	Cameron
PyTorch MLP	Model 2 (deep learning)	BCEWithLogitsLoss; class weighting tuned: pos_weight computed as for XGBoost (≈ 13.4, estimate) or 1; AdamW; early stopping on inner split PR AUC; final refit at median best epoch; 30 random configurations; final model trained on the same 5 seeds	Zack

Search spaces. XGBoost: class weighting on or off; max_depth 3 to 6; learning_rate 0.01, 0.03, 0.1; up to 2,000 trees; subsample 0.7, 0.85, 1.0; colsample_bytree 0.5, 0.7, 1.0; min_child_weight 1, 5, 10. MLP: class weighting on or off; 1 to 3 hidden layers; width 64, 128, 256; dropout 0.1, 0.3, 0.5; learning rate 0.0001, 0.0003, 0.001; weight decay 0.00001 to 0.001; batch 256; up to 300 epochs; patience 20.

Calibration and thresholds. Every scorer is calibrated with Platt scaling, fixed in advance. Each calibrator is fitted on the concatenated out of fold predictions from all three CV repeats of the final configuration; for original Z′, the out of fold predictions are the scores themselves. Cost thresholds for each cost ratio are chosen on the same out of fold predictions, never on the test set. Rank metrics (PR AUC, ROC AUC) use uncalibrated scores.

Seeds. XGBoost and MLP final models are each trained on the same 5 seeds from `config.yaml`. All headline metrics, out of fold predictions, calibrators, thresholds and paired tests use the seed 42 model. The 5 seed mean and standard deviation of PR AUC are reported for both models as stability evidence only.

Metrics. Primary: PR AUC. Secondary: ROC AUC, Brier score with reliability diagrams, recall and precision at a 5% false positive rate. Cost analysis: expected cost at the out of fold thresholds for miss to false alarm cost ratios of 5, 10, 20 and 50. Significance: paired

bootstrap of PR AUC differences for H1a to H1c and H2 (rules in Section 1); DeLong test for ROC AUC. Horizon comparison (H3): ROC AUC as the decision metric, PR AUC divided by test prevalence as supporting. Explanation: `explain.py` runs SHAP on training data only, on the frozen models, on Wed 25 to Thu 26 Nov straight after the freeze, and compares the drivers with the five Altman inputs.

Experiments

ID	Experiment	Answers	If time runs short
E1	All five scorers on the 1 year horizon file (`5year.arff`), frozen split	H1a to H1c, H2	Never cut
E2	Frozen XGBoost and MLP configurations (fixed iteration counts, seed 42, no fresh tuning) on all five files. `5year.arff` reuses the frozen main split and the E1 test results and is not scored a second time; each other file gets its own stratified 80/20 split (seed 42) after duplicate removal, with preprocessing refitted on that training split. Rank metrics only, no calibration or cost analysis. H3 decided on `5year.arff` versus `1year.arff` ; the five file trend is described, not tested	H3	Reduced to `1year.arff` second; cut third
E3	XGBoost native missing handling versus imputation; feature selection on versus off	Preprocessing justification	Cut first

Environment. Python 3.11; pandas, numpy, scipy, sklearn, xgboost, torch, shap, matplotlib, pinned in `requirements.txt` . Runs on a laptop CPU.

5. Roles and workload

Technical work stays equal, now at 18 planning hours each after revision 3; admin and project setup sit with Abdalla on top of that.

- **Abdalla:** project lead and final decisions; project base setup; pull request reviews; test run; submission. Technical: three baselines (original Z′, refitted Z′, logistic regression), evaluation harness and hypothesis tests, SHAP, H3 analysis.
- **Cameron:** deputy lead with merge rights, authorised to execute the test run procedure and the submission checklist if Abdalla is unavailable (handover recorded in the decision log). Data and Model 1 lead: EDA, preprocessing and feature selection, XGBoost and its 5 seed runs, E2 XGBoost runs, E3 ablation, error analysis.
- **Zack:** deep learning lead. PyTorch MLP, tuning, 5 seed runs and the stability summary for both models, E2 MLP runs, Platt calibration for all five scorers, Figures 2 and 3 and their Section 8 paragraphs.

Technical workload (planning estimates; admin and setup excluded)

Member	Technical components (estimated hours)	Technical (hours)	Writing (hours, 2 per counted page)	Combined (hours)
Abdalla	Original Z′ on pipeline output (1.5); refitted Z′ with standardised inputs and C tuning (1.5); logistic regression on 64 ratios with C tuning (2); evaluation harness: stub interface, stratified shared bootstrap, six H1 comparisons, H2 verdicts, DeLong, out of fold cost thresholds (6); SHAP on training data for both frozen models (3); H3 test and horizon analysis (2); test run and results tables (2)	18	5.8 (2.9 pages)	23.8

Member	Technical components (estimated hours)	Technical (hours)	Writing (hours, 2 per counted page)	Combined (hours)
Cameron	EDA notebook (4); preprocessing pipeline (4.5); feature selection with Altman inputs protected (1.75); XGBoost with tuned class weighting, inner split early stopping, fixed count refit (4.25); XGBoost final runs on 5 seeds (1); E2 XGBoost runs on the four other files (1); E3 ablation and error analysis (1.5)	18	4.6 (2.3 pages)	22.6
Zack	MLP architecture and dataloader (3); training loop, inner split early stopping, seed control, fixed epoch refit (4); 30 configuration search with tuned class weighting (3); final 5 seed runs and stability summary for both models (1.5); Platt calibration for all five scorers on concatenated out of fold predictions, reliability diagrams (3); PR and cost curve figures (2.5); E2 MLP runs on the four other files (1)	18	3.2 (1.6 pages)	21.2

Technical hours: 18 each (17 in revision 2). Revision 3 adds about 3 technical hours in total; to keep the split equal, the E2 MLP runs move from Abdalla to Zack and Zack takes the 5 seed stability summary for both models. **Writing check:** the largest combined gap is 2.6 hours (Abdalla over Zack), under the 3 hour threshold, so no member is flagged. If anyone's technical load runs more than about 3 hours over, a task is moved at the Sunday meeting; if the whole group runs short of time, the cut order in Section 6 applies.

Effort table categories

Category	Abdalla	Cameron	Zack
1. Data and preprocessing	Dataset decision; project base (data, loader with duplicate removal, frozen split, config); Altman variable mapping; leakage review	**Lead:** EDA, infinite and missing values, winsorisation, correlation filter with Altman inputs protected	PyTorch dataloader; MLP scaling choice; EDA figures on ratio skew
2. Modelling	Original Z′ on pipeline output; refitted Z′ and logistic regression (standardised, C tuned)	**Lead:** XGBoost, search, 5 seed runs, E2 XGBoost runs, E3 ablation	**Lead:** MLP architecture, training loop, search, 5 seed runs, E2 MLP runs
3. Evaluation	**Lead:** harness (stratified shared bootstrap), H1a to H1c, H2 verdicts, H3 test, cost thresholds, test run, SHAP on training data	XGBoost CV results; E3 results; error analysis	MLP CV results; Platt calibration for all five scorers; 5 seed stability summary; PR and cost figures
4. Report	Abstract, background, evaluation strategy, Section 8 results table and hypothesis verdicts, conclusions, final edit, references	EDA, preprocessing, Model 1	Model 2; Section 8 paragraphs on Figures 2 and 3

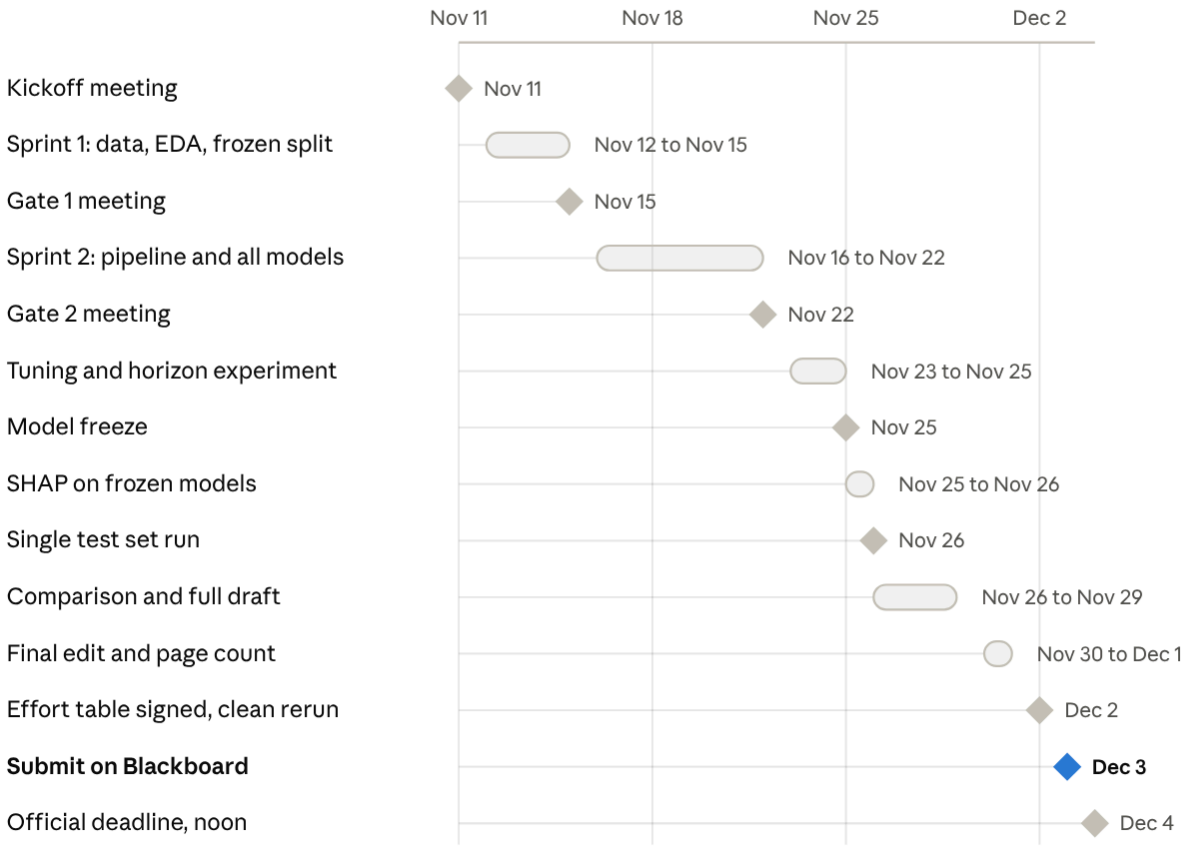
All three sign 100% in every category on Wed 2 Dec. Commits, pull requests and the task board are the evidence of contribution.

6. Timeline

Two hard gates (Sun 15 and Sun 22 Nov), a model freeze on Wed 25 Nov, and submission on Thu 3 Dec.

Submission lands a day before the 4 December deadline

CS3AM plan by week, 2026; diamonds are milestones, bars are work periods



CS3AM plan · 11 Nov to 4 Dec 2026

Gate	Date	Must be true
Gate 1	Sun 15 Nov	Project base live with duplicate removal on all five files; split frozen; stub `evaluate.py` interface merged; hypothesis rules, margins, seed 42 and the Platt choice committed to the decision log; Cameron's Blackboard submission access confirmed; EDA notebook merged; Altman mapping verified; first MLP training loop runs
Internal milestone	Tue 17 Nov	Preprocessing pipeline merged (Cameron)
Gate 2	Sun 22 Nov	All five scorers produce out of fold predictions (all three CV repeats) and CV metrics through the shared harness; report sections 3 to 5 drafted

Gate	Date	Must be true
Model freeze	Wed 25 Nov	No further tuning; final configurations, fixed iteration counts, 5 seed final models, Platt calibrators and out of fold cost thresholds committed
SHAP run	Wed 25 to Thu 26 Nov	SHAP on training data for the frozen XGBoost and MLP complete
Full draft	Sun 29 Nov	Every section drafted; all figures produced

If a gate is missed, scope is cut in this order: (1) E3; (2) E2 reduced to `1year.arff`; (3) E2 cut entirely; (4) MLP SHAP. E1 with all five scorers, Platt calibration and out of fold thresholds are never cut. The dates do not move.

7. Ways of working and delivery

One meeting a week, daily updates on WhatsApp, and every change reaches `main` through a pull request Abdalla approves.

- **Meetings:** Sundays, 45 minutes (15, 22 and 29 Nov): gate review, results, plan for the week.
- **WhatsApp group:** daily update by 21:00 in three lines (done, next, blocked). Blockers raised the same day. Replies within 24 hours.
- **Repository:** private GitHub `cs3am_bankruptcy`, set up by Abdalla. Branches named `owner/topic`; small pull requests; reviews within 24 hours.
- **Deputy:** Cameron, with merge rights. He reviews and merges when Abdalla has not reviewed within 24 hours or is unavailable. He is also authorised to execute the test run procedure and the submission checklist if Abdalla is unavailable, with the handover recorded in the decision log. Otherwise, changes to `split.py`, the test indices and `evaluate.py` need Abdalla.
- **Task board:** GitHub Projects (To do, Doing, Review, Done).
- **Hours log:** shared sheet, updated weekly; feeds the front page.
- **Decisions:** discussed in the group, decided by Abdalla, recorded in `docs/decision_log.md`.

Repository layout

```
cs3am_bankruptcy/
  README.md, requirements.txt, config.yaml
  data/raw/            five ARFF files
  data/splits/         frozen test indices (Abdalla, or Cameron as authorised
deputy)
  src/load.py          src/split.py        src/preprocess.py
  src/baselines.py     src/xgb_model.py    src/mlp_model.py
  src/evaluate.py      src/explain.py
  notebooks/01_eda.ipynb  02_models.ipynb  03_results.ipynb
  results/             metrics (JSON), figures (PNG, 300 dpi)
  docs/decision_log.md
```

Code standards

- PEP 8 with `black`; docstrings and type hints on every function in `src/`.
- Notebooks import from `src/`; no logic lives only in a notebook.
- Seeds and paths come from `config.yaml`, never hard coded.
- Borrowed code carries a source comment.
- Every member can explain every line they commit, including GenAI assisted code.
- Only `evaluate.py` may read the test indices.

Definition of done: merged, runs from a clean environment, results saved to `results/`, figure saved, report paragraph drafted.

Test run procedure

1. Wed 25 Nov, model freeze: final configurations, fixed iteration counts, 5 seed final models, Platt calibrators and out of fold cost thresholds are committed and the commit is tagged `model_freeze`. SHAP then runs on training data only (Wed 25 to Thu 26 Nov).
2. Thu 26 Nov: Abdalla runs `evaluate.py` once, from a clean environment at that tag, on the frozen `5year.arff` test indices and the four other E2 test splits. If Abdalla is unavailable, Cameron runs it as authorised deputy, with the handover recorded in the decision log. The other confirms the commit hash.
3. Test predictions, metrics JSON and figures are committed unchanged; the commit hash goes in the decision log.
4. No tuning after the run. If a genuine bug is found, the fix and the rerun are logged and disclosed in the report.

Submission checklist

- [ ] Report: at most 7 counted pages, Times New Roman 12 pt, 1.15 spacing, at most 4 figures in counted pages.

- [ ] Front page: module code, title, date, actual hours, other learning aids (GenAI listed).
- [ ] Effort table signed by all three, 100% in every category (Wed 2 Dec).
- [ ] References checked against DOIs; Harvard style consistent.
- [ ] Clean environment rerun of `03_results.ipynb` reproduces the reported numbers (Wed 2 Dec).
- [ ] Zip contains the report, `data/raw` (five ARFF files, CC BY 4.0 credit in the README), `src/`, `notebooks/`, `requirements.txt`, `config.yaml`, README and `results/`.
- [ ] Borrowed code carries source comments.
- [ ] Abdalla (or Cameron as authorised deputy, handover logged) submits once on Thu 3 Dec, downloads the submitted zip to check it opens, and posts the Blackboard receipt in the WhatsApp group.

8. Report plan

The report must fit 6.8 counted pages, leaving 0.2 pages under the 7 page limit; extra tables and plots go to appendices, which are not counted.

No.	Section	Marks	Pages	Author	Reviewer
1	Abstract	Report quality	0.2	Abdalla	Zack
2	Background and problem	5	0.5	Abdalla	Cameron
3	Exploratory data analysis (Figure 1)	5	0.8	Cameron	Abdalla
4	Preprocessing and feature selection	10	0.7	Cameron	Abdalla
5	Model 1: XGBoost (approach and training)	Shared 30 for modelling	0.7	Cameron	Zack
6	Model 2: PyTorch MLP (approach and training)	as above	0.7	Zack	Cameron
7	Performance measures and evaluation strategy	10	0.6	Abdalla	Zack
8	Results comparison: Figures 2 and 3 with their paragraphs; the five scorer results table with the hypothesis verdicts; one sentence with the headline H3 numbers	5, plus 5 for results	1.4 (Zack 0.8, Abdalla 0.6)	Zack (Figures 2 and 3 and their paragraphs); Abdalla (results table, hypothesis verdicts, H3 sentence)	Cameron

No.	Section	Marks	Pages	Author	Reviewer
9	Conclusions, recommendations, limitations, future work (Figure 4)	10	0.9	Abdalla	All
10	Individual reflection	5	0.3	One paragraph each (0.1 each)	Abdalla
Not counted	Front page, signed effort table, references, appendices	Report quality	0	Abdalla	All

Figures in the counted pages (at most 4)

1. Class balance and missingness by ratio (Section 3).
2. Precision recall curves, all five scorers, test set (Section 8, headline figure).
3. Expected cost against cost ratio, all five scorers, thresholds from out of fold predictions (Section 8).
4. SHAP top 10 drivers, XGBoost and MLP side by side, against the Altman inputs (Section 9).

Appendix (not counted): the five Altman inputs by class after winsorising; MLP training and validation curves; reliability diagrams for all five scorers; the E2 horizon results table (rank metrics only); the 5 seed stability table for XGBoost and the MLP; full search grids and metric tables. Per model results sit in the Section 8 table, so Sections 5 and 6 carry approach and training only.

Format: Word on OneDrive, Times New Roman 12 pt, 1.15 spacing, Harvard references via Zotero.

9. Risks and controls

Schedule slip and test set leakage are the two risks most likely to cost marks.

Risk	Control	Owner
Schedule slip (technical scope up about 9 hours since revision 1)	Hard gates; model freeze 25 Nov; cut order in Section 6	Abdalla

Risk	Control	Owner
Test set leakage	Transforms fitted inside folds; SHAP on training data only; test read only by `evaluate.py`; one test run (Section 7 procedure); the `5year.arff` test set is not scored again in E2	Abdalla
Calibration or threshold leakage	Platt calibrators and cost thresholds fitted on out of fold training predictions only	Zack, Abdalla
Early stopping leakage	Inner split of the training fold; final refit at median best iteration from CV	Cameron, Zack
Post hoc changes to hypothesis rules	Rules, margins, seed 42 and the Platt choice committed to the decision log by Gate 1; not altered after the test run	Abdalla
Duplicate rows inflating results	`load.py` removes exact duplicates from all five files before any split; counts logged and reported	Abdalla
Plan figures drifting from the data	Post duplicate counts and class weights computed and logged by the code; plan figures marked as estimates	Abdalla
Page limit overrun	Page budget per section; at most 4 counted figures; detail to appendices	Abdalla
MLP unstable or clearly worse	Both models trained on 5 seeds; headline uses seed 42; stability reported; H2 judged on a 0.05 margin fixed in advance	Zack
Unequal contribution or workload drift (18 technical hours each)	Visible task board; technical and writing hours checked each Sunday	Abdalla

Risk	Control	Owner
Abdalla or another member unavailable	Cameron authorised to run the test run and submission (handover logged); Blackboard access confirmed at Gate 1; Cameron and Zack review each other's code; self certification extension available	All
Tuning on `5year.arff` favours that file and may overstate decay	Stated in the report's limitations	Abdalla
Sampling period bias in the data	Stated and discussed as a limitation	Abdalla
Code fails on a clean machine	Pinned requirements; clean rerun on 2 Dec	Abdalla

10. Actions to Gate 1 (Sun 15 Nov)

Gate 1 passes only if the project base is live, the split is frozen, the stub `evaluate.py` interface is merged, the fixed in advance rules are committed, Cameron's Blackboard access is confirmed, the EDA notebook is merged and the first MLP training loop runs.

Abdalla

- [ ] Project base live: repository, task board, `requirements.txt`, `config.yaml` (including the 5 seeds and the bootstrap seed), raw data, `load.py` with row count checks and exact duplicate removal on all five files (counts logged); both members can clone and run it (Fri 13 Nov).
- [ ] `split.py` and frozen test indices committed (Fri 13 Nov).
- [ ] Stub `evaluate.py` interface merged: input format for out of fold and test predictions (ids, labels, scorer name) and output format (metrics JSON) (Sun 15 Nov).
- [ ] Hypothesis rules, margins, seed 42 and the Platt choice committed to `docs/decision_log.md` (Sun 15 Nov).
- [ ] Confirm Cameron can submit on Blackboard (Sun 15 Nov).
- [ ] Z′ coefficients and zones verified against Altman (1983); `baselines.py` with original Z′ on pipeline output, refitted Z′ and logistic regression (standardised, C tuned) (Sun 15 Nov).
- [ ] Reading notes: Altman (1983), Ohlson (1980), Saito and Rehmsmeier (2015), DeLong et al. (1988), Lundberg and Lee (2017).

Cameron

- [ ] `01_eda.ipynb`: class balance, missingness by ratio, infinite values, distributions, correlation heatmap, the five Altman inputs by class (Sun 15 Nov).
- [ ] EDA section drafted, about 1 page (Sun 15 Nov).
- [ ] Preprocessing pipeline merged, with the Altman inputs protected in the correlation filter (Tue 17 Nov, internal milestone).
- [ ] Reading notes: Altman (1968), Zięba et al. (2016), Barboza et al. (2017), Chen and Guestrin (2016).

Zack

- [ ] First `mlp_model.py`: dataset class, training loop, early stopping, seed control, tested on a 1,000 row sample (Sun 15 Nov).
- [ ] EDA figures on ratio skew to inform the MLP scaling choice (Sun 15 Nov).
- [ ] Reading notes: Grinsztajn et al. (2022), Loshchilov and Hutter (2019), Srivastava et al. (2014), Guo et al. (2017).

All

- [ ] Start the hours log.
- [ ] Daily WhatsApp update by 21:00.

Reading notes are 150 words per source: what it claims, its method, and the report sentence it supports.

11. References

Dataset facts were taken from the UCI page on 11 Nov 2026; literature details are to be checked against DOIs before they enter the report.

- Polish Companies Bankruptcy, UCI Machine Learning Repository. Tomczak, S. (2016). DOI 10.24432/C5F600. CC BY 4.0.

Source	Publication
Altman (1968)	*Journal of Finance*, 23(4), 589 to 609
Altman (1983)	*Corporate Financial Distress*, Wiley
Barboza, Kimura and Altman (2017)	*Expert Systems with Applications*, 83, 405 to 417
Chen and Guestrin (2016)	Proceedings of KDD 2016, 785 to 794
DeLong et al. (1988)	*Biometrics*, 44(3), 837 to 845
Grinsztajn, Oyallon and Varoquaux (2022)	NeurIPS 35, Datasets and Benchmarks track

Source	Publication
Guo et al. (2017)	Proceedings of ICML 2017
Loshchilov and Hutter (2019)	Proceedings of ICLR 2019
Lundberg and Lee (2017)	NeurIPS 30
Ohlson (1980)	*Journal of Accounting Research*, 18(1), 109 to 131
Saito and Rehmsmeier (2015)	*PLoS ONE*, 10(3), e0118432
Srivastava et al. (2014)	*Journal of Machine Learning Research*, 15, 1929 to 1958
Zięba, Tomczak and Tomczak (2016)	*Expert Systems with Applications*, 58, 93 to 101

12. Change log

Revision 3 (11 Nov 2026)

1. Calibration: Platt only for every scorer, fitted on concatenated out of fold predictions from all three CV repeats; isotonic, the Brier choice and its cut step removed.
2. Both logistic regressions standardised, L2, class weight balanced, C tuned over 0.01 to 10; original Z′ computed on pipeline output so every row has a score.
3. H1 split into H1a to H1c (six comparisons, all intervals reported, no pooled verdict, no correction); H2 margin 0.05 with ordered verdicts.
4. Both models trained on 5 seeds; the seed 42 model drives all headline results; bootstrap stratified with shared indices and a configured seed.
5. H3 decided on ROC AUC via independent stratified bootstraps of the `5year` and `1year` test sets; E2 reuses the `5year` split and E1 results, rank metrics only; tuning bias added to limitations.
6. Hypothesis rules, margins, seed 42 and the Platt choice committed to the decision log by Gate 1.
7. Cameron authorised to run the test run and submission (handover logged); Gate 1 adds a Blackboard access check.
8. SHAP runs on training data only, Wed 25 to Thu 26 Nov; timeline row added.
9. Section 8 split (Zack 0.8, Abdalla 0.6 pages); writing hours added; technical hours 18 each; no writing flag (largest gap 2.6 hours).
10. Dataset counts labelled raw; `2year` to `4year` marked Horizon experiment (H3); 13.4 and split sizes marked as estimates.

Revision 2 (11 Nov 2026)

1. Baselines: refitted Z′ (logistic regression on the five Altman inputs, Abdalla) added; RQ1, H1, results and pipeline diagram reframed around five scorers.
2. H3 and RQ3 judged on ROC AUC and PR AUC divided by prevalence; E2 runs frozen XGBoost and MLP configurations on all five files, own 80/20 split each (seed 42), no retuning.
3. Calibration (Platt or isotonic, chosen on out of fold Brier) for every scorer including Z′; cost thresholds set on the same out of fold predictions; class weighting tuned in both searches.
4. Early stopping on a 15% inner split of the training fold; final refit at the median best iteration from CV.
5. Correlation filter never drops Attr3, Attr6, Attr7, Attr8 or Attr9.
6. H1 passes if the paired bootstrap 95% interval excludes zero; H2 uses a 0.02 PR AUC margin fixed in advance.
7. Cameron named deputy with merge rights; test run procedure and submission checklist added; Gate 1 adds the stub `evaluate.py` (Abdalla, 15 Nov) and merged pipeline (Cameron, 17 Nov).
8. Report: 4 counted figures (Altman inputs and MLP curves to appendix, SHAP to Section 9); Section 8 raised to 1.4 pages; total stays 6.8.
9. `load.py` removes exact duplicate rows before the split and logs the count.
10. Workload 17 hours each (E2 XGBoost runs moved to Cameron); all three flagged at +2 hours; risk table and cut order updated.

13. Conflicts found in the revision 3 instructions

Six instructions sit uneasily with the existing plan or with each other; each was applied as written or as noted below, and none was resolved silently.

1. **Items 1 and 5:** item 5 says out of fold predictions use the seed 42 model, but out of fold predictions come from the CV models, not the final model. Applied as: the CV runs of the final configuration use seed 42. Needs confirmation.
2. **Item 2 and the benchmark definition:** computing original Z′ on imputed, winsorised inputs means it is no longer exactly the published score for extreme or incomplete firms, and Altman′s zones (1.23 and 2.90) hold only approximately. Applied as instructed; the report must state this.
3. **Items 4 and 8:** the 0.05 margin is justified by an expected interval width of about ±0.05, but the margin must be fixed by Gate 1 (Sun 15 Nov), before any out of fold predictions exist to check that width (Gate 2, Sun 22 Nov). Applied as instructed; the width is an estimate.

4. **Items 7 and 12:** item 12 labels `2year` to `4year` as "Horizon experiment (H3)", but item 7 tests H3 on `5year` versus `1year` only and describes the other files without testing them; the cut order also drops them first. Labels applied; those files support description only.
5. **Item 10 and the schedule:** SHAP (Wed 25 to Thu 26 Nov) overlaps the test run (Thu 26 Nov), both owned by Abdalla, and MLP SHAP sits last in the cut order but now runs before the full draft, so it can only be cut before 25 Nov. Kept as instructed.
6. **Item 9 and the revision 2 deputy rule:** item 9 updates the wording for `data/splits` and `evaluate.py` but not `split.py`. Applied as: outside the test run procedure, changes to `split.py` still need Abdalla.